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To the Intel AI Engineering Team,

The job description puts it exactly right: the model is only half an agent; the harness is the other half. I have spent the last year building the harness half — the agent loop, tool integrations, MCP servers, and multi-model abstraction layers that make agentic systems actually work in production. That work maps directly to every deliverable in this role.

- **MCP Servers and Tool Integrations, Shipped in Production:** At **Isomer AI**, I built MCP servers in Python and TypeScript that expose customer-internal tools to LLM agents — file I/O, browser automation via Playwright, Tailscale VPN tunneling for secure internal access. This is the exact “tool integrations” mandate: file, terminal, web, and message app connectors behind a clean abstraction.
- **Model-Agnostic Multi-LLM Routing:** At Isomer and through **KubeServe**, I built model-agnostic routing layers that swap local and cloud providers behind one API — Claude, GPT-4o, and Gemini, with circuit breaker and EWMA load balancing. The role’s requirement to keep the harness “model-agnostic so local and cloud models are swappable behind one abstraction” is architecture I have already implemented.
- **Agent Loop with Temporal.io Orchestration:** **AlgoPulse** is a production agentic AI platform built on Temporal.io and Kafka — multi-step reasoning pipelines, tool invocation, retry logic, and termination conditions at 2,000 async inference requests per second. I have designed and debugged the agent loop that this role builds at Intel’s scale.
- **Observability and Reproducible Sessions:** Through **LLM Eval Harness** and **KubeServe**, I instrumented agent pipelines with OpenTelemetry, Prometheus, and Grafana — tracing individual tool calls, replaying evaluation runs against frozen test sets, and publishing reproducible performance comparisons across 6 metrics. Making agents observable and debuggable is not an afterthought in my work; it is the foundation.

Intel’s hybrid local–cloud agent architecture is the right technical direction: private, efficient, and cost-predictable by design. I want to help build the harness that makes it real. I would be glad to discuss.

Sincerely,

Aditya Kulkarni